

Effects of female access and diet on insemination success, senescence, and the cost of reproduction in male Mexican fruit flies (*Anastrepha ludens*)

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Effects of diet and mating on *A. ludens* male fitness traits

Abstract:

Hypotheses regarding the life history trade-off between survival and reproduction have been extensively tested in female insects, but have been rarely explored in males. To test the effect of dietary conditions on male life history trade-offs between survival and reproduction, measured changes in the life expectancy and age-specific reproductive ability of male Mexican fruit flies, *Anastrepha ludens* (Diptera: Tephritidae) were measured, resulting from delayed access to protein and mates. There was a clear cost of female access for males with access to dietary protein, measurable as a decrease in life expectancy, which was further influenced by the age when females were introduced. A protein deficient diet reduced the lifespan benefit of virginity and masked the detrimental effect of female access on male life expectancy. Dietary protein was not necessary for reproductive success, but access to protein at eclosion improved the lifetime reproductive success of males compared to when it was delayed. Overall, reproductive success diminished as the males aged, regardless of the dietary conditions, providing evidence for reproductive senescence in males. Delaying the males' access to a protein source failed to influence the negative effect of age on reproductive ability. Since age specific reproductive rates declined with age, regardless of diet, male fitness did not benefit from extending lifespan. Therefore, males are expected to be allocating available resources towards reproductive effort in favour of extended lifespan, regardless of mate and protein availability.

Keywords: Male reproduction, male aging, cost of mating, trade-offs, resource allocation, Tephritidae

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39 **Introduction**

40 When nutritional scarcity is encountered in an environment, reduced rates of senescence
41 are often observed in animals, resulting in an extension in lifespan while reproductive effort is
42 reduced or arrested (Weithoff, 2007; Carey *et al.*, 2008). By avoiding the cost of reproduction
43 and allocating available nutrients to somatic upkeep during unfavourable dietary conditions,
44 organisms may improve their fitness by surviving until successful reproduction can begin or
45 resume (Wiley, 1974; Tuljapurkar, 1990; Barrett *et al.*, 2009). Although there is a deep literature
46 on the physiological response to dearth periods, most research conducted on the factors affecting
47 reproductive trade-offs in insects primarily focuses on females in response to dietary restriction
48 and the timing of egg production (Awmack & Leather, 2002; Boggs & Freeman, 2005; Carey, *et*
49 *al.*, 2008; Harwood *et al.*, 2013). In contrast, studies investigating reproductive trade-offs in male
50 insects are uncommon, with research primarily focused on the cost of reproduction in the
51 absence of interactions with nutritional conditions (Partridge & Farquhar, 1981; Kotiaho &
52 Simmons, 2003; Pauku & Kotiaho, 2005; Papadopoulos *et al.*, 2010).

53 In male insects, reproductive costs have been linked to many of the sexual behaviours
54 and physiological processes related to courtship, signalling, mating and insemination (Martin &
55 Hosken, 2004; Kuijper *et al.*, 2006; Papadopoulos, *et al.*, 2010). Accordingly, male reproductive
56 effort is expected to be postponed when the likelihood of successful reproduction is low, thereby
57 preventing the cost of reproduction until the probability of reproductive success improves (Perrin
58 & Sibly, 1993). Such a deferment in the cost of reproduction has been well documented in the
59 females of several species in the family Tephritidae (Carey, 2003). Specifically, female tephritids
60 will enter a waiting period characterized by little or no reproductive effort and a lowered risk of

mortality during unfavourable reproductive conditions. When conditions improve, the waiting period ends and the females experience an increase in age specific reproductive ability and a reduced risk of mortality (Carey *et al.*, 1998).

In male tephritids, the lack of female access alone will not elicit a comparable waiting period, since males will still form leks and produce and release sexual attractant pheromones in the absence of females, each of which potentially contribute to the cost of reproduction (Liedo *et al.*, 2002; Johansson *et al.*, 2005; Zhang *et al.*, 2006). A source of dietary protein appears to be essential for maximizing male tephritid reproductive ability, with copulation frequency, sexual signalling rate, pheromone production, and participation in leks all dependent on diet (Aluja *et al.*, 2001; Yuval *et al.*, 2002; Liedo *et al.*, 2013). Therefore, the absence of certain dietary resources appears to prevent or reduce the reproductive effort of males. This reduced reproductive effort due to poor dietary conditions is expected to allow males to avoid the cost of reproduction and allow nutritional resources to be used for somatic upkeep, therefore improving life expectancy, especially when females are not available. Since an extended lifespan is assumed to only be adaptive if it results in greater reproductive success later in life (Bonduriansky *et al.* 2008), such an improvement in lifespan is expected to result in greater reproductive potential at advanced ages. Contrary results to this hypothesis have been presented in the males of some dipteran species, in which male reproductive rates and attractiveness have been observed to decrease with age (Bonduriansky & Brassil, 2005; Papanastasiou *et al.*, 2011), so that lifespan extension cannot be expected to improve male fitness.

Accordingly, the effects of dietary conditions and female availability on the longevity and age specific insemination success of the male Mexican fruit fly (*Anastrepha ludens*) were tested to determine if male reproductive effort could be delayed, thus extending male lifespan

and slow reproductive senescence, if observed . The Mexican fruit fly, a significant economic pest (Dominguez *et al.*, 2010), has been shown to be a useful model organism for understanding demographic trade-offs (Carey, *et al.*, 2008; Carey & Molleman, 2010). Much of what is known about the reproductive behaviour and life history of the male Mexican fruit fly, has been derived from studies focused on its effectiveness in the sterile insect technique (SIT) (Shaw *et al.*, 1967; Robacker & Garcia, 1993). These studies rarely focus on the trade-offs between reproduction and survival in males, and male reproductive ability has not been observed through the entire lifespan (Aluja *et al.*, 2000; Aluja *et al.*, 2008). Therefore, the data presented here, regarding male Mexican fruit fly life history characteristics, will be useful in controlling their populations and inform hypotheses regarding life history trade-offs in male insects.

Materials and Methods

Husbandry

Flies were obtained from the Moscafrut mass-production facility at Metapa de Domínguez, Chiapas, Mexico, which were reared using techniques described by Dominguez *et al.* (2010). When these experiments were conducted the colony was approximately 170 generations old. Wild flies have been introduced into this colony periodically to maintain genetic diversity. All experiments were conducted under laboratory conditions (LD 12:12 L: D at $23 \pm 3^{\circ}$ C and 60-75 % relative humidity). Adults were separated by sex, and 450 males were individually housed in $4 \times 4 \times 10$ cm Plexiglas cages from eclosion until death. Treatments consisted of combinations of dietary conditions and female availability, with 50 males per treatment. All flies were provided daily access to water and to one of two *ad libitum* solid diets: 1) a full diet, which included a carbohydrate and a protein source (3:1 ratio mixture of sugar to yeast hydrolysate) or 2) a diet consisting only of a carbohydrate (sugar only diet). The

experimental control consisted of 50 males provided *ad libitum* solid full diet from eclosion, adult age 0 days, until death and never paired with virgin females.

The difference in life expectancy of males with female access compared to those in the control was used as a proxy for the cost of reproduction. Cohorts of reproductive virgin females (age 10-15 days) were housed in large cages with several hundred same aged females and provided with constant access to water and full diet. Cohorts of virgin females were maintained for the duration of the study and resupplied with new females weekly to maintain the availability of 8-15 day olds. In the appropriate treatments, a single female was introduced into each of the males' cage. After 24 hours of access, the female was removed and a new virgin female was presented.

Diet and Female Access

Three treatments were used to establish the cost of reproduction and the influence of diet type on male longevity: 1a) full diet from eclosion until death with constant female access, in which males received access to a new virgin female daily for their entire lifespan; 2a) sugar diet from eclosion until death with constant access to virgin females for the entire lifespan; or 3a) sugar-only diet with no female access. The cost of female access and the effect of diet quality on male lifespan were estimated by comparing the life expectancy of the males in each treatment with that of males in the control.

Truncated Reproduction

To describe the influence of prior female access on male lifespan after females were no longer available, 50 males were assigned to two truncated access treatments: 1b) constant access to full diet for the entire lifespan, with virgin females only provided from eclosion until age 20 days, or 2b) sugar diet for the entire lifespan, with virgin females only provided until age 20

days. These two treatments were compared to those testing the effect of constant female access of males provided both diets and to those that remained unpaired.

Delayed Reproduction

In an attempt to delay reproduction in males, access to virgin females, to full diet, and to both females and full diet were delayed for 20 days post eclosion. Fifty males were tested under each of the following delayed reproduction treatments: 1c) constant access to full diet from eclosion until death, while access to virgin females was delayed for 20 days, after which females were provided daily until death; 2c) sugar diet provided until age 20 days, after which full diet was provided until death, while daily virgin females were provided daily from eclosion until death; or 3c) both full diet and virgin females delayed until age 21 days, prior to which males were given sugar diet and not provided females. The treatment in which full diet and female access was constant for the entire lifespan (Treatment 1a from *Diet and Female Access*) was included for a factorial design to determine if delayed diet or female access improved the males life expectancy compared to when reproduction was expected to be continuous.

Diet and Age Specific Reproduction

To quantify the effect of age and dietary conditions on male lifetime reproductive success, the insemination success twenty males from the treatments 1a, 2a, and 2c were monitored every fifth day beginning at eclosion. The females associated with these males on the observation days were removed after being paired and then held for 15 days in individual cages with an egg laying substrate (silicon coated mesh) and access to water and full diet. Eggs were collected and observed for larval hatch over an additional 15 day period. The presence of larvae indicated successful insemination for the male at the age when it was paired with the female, while a lack of larval hatch was interpreted as failed insemination. Daily insemination success

was recorded as the proportion of males associated with larval production at each observation period.

Statistical Methods

The effect of diet and female access on male life expectancy was tested with two linear regressions. The first assumed the control as the baseline for estimating the effects of constant diet, constant female access, and truncated female access on the male lifespan. The second assessed the effects of delaying female access, full diet, and both on lifespan compared to the treatment 1a. This treatment was selected as the baseline to determine if delaying access to females and/or full diet would lead to lifespan extension compared to when reproduction was assumed to be constant. Additional pair wise comparisons were made among treatments with two-tailed t-tests. A final linear regression model analysed the effects and interaction of diet type and age on male insemination success, with the treatment 1a selected as the intercept and age was analysed as a covariate.

Results

Diet and Female Access

The life expectancy of males fed full diet and never paired with females was 70.32 days. Female access reduced the life expectancy of males by 20.74 days ($t = -5.11$, $P < 0.001$), but female access was not observed to reduce the probability of survival until after age 20 days (Fig. 1 and 2). The life expectancy of males on sugar diet without female access was reduced by 26.94 days compared to when full diet was provided ($t = -6.64$, $P < 0.001$; Figure 2). When sugar diet was combined with constant female access the male life expectancy was 25.04 days less than that of control flies ($t = -3.95$, $P < 0.001$), but was not different from the life expectancy of males supplied full diet and constant female access ($t = -1.31$, $P = 0.20$; Fig. 2). There was no difference

in the life expectancies of sugar fed males with and without female access ($t = -0.61$, $P = 0.54$; Fig. 2), and age specific survival was similar in both of these treatments (Fig. 1).

Truncated Female Access

Truncated female access combined with full diet did not reduce the life expectancy of males relative to the control ($t = -1.84$, $P = 0.07$), and extended the lifespan relative to those that had constant female access by 13.10 days ($t = 3.27$, $P = 0.001$; Fig. 2). In combination with sugar diet, the truncated female access resulted in a significant loss of life expectancy compared to the control ($t = 1.75$, $P = 0.01$), but lessened the loss of life expectancy associated with the sugar diet by 7.3 days ($t = -2.15$, $P = 0.03$), and appeared to improve age specific survival after age 40 days (Fig. 1). The early exposure to females did not prevent the cost of female access while males were fed sugar diet ($t = -1.40$, $P = 0.16$; Fig. 2).

Delayed Reproduction

When full diet was always provided, delaying female access reduced the life expectancy of males by 11.60 days compared to the control ($t = -2.87$, $P = 0.004$), but improved the life expectancy of males by 9.20 days compared to when female access was constant ($t = 2.38$, $p = 0.020$; Fig. 3). When female access was constant from eclosion until death, but full diet was delayed for 20 days, the life expectancy of males was shown to be no different from when full diet and female access were constantly provided ($t = 1.30$, $P = 0.199$). Additionally, the lifespan benefit of delayed female access was lost when combined with delayed full diet, so that males with delayed female access and delayed full diet had the same life expectancy as males with constant access to both ($t = -1.96$, $P = 0.107$; Fig. 3).

Diet and Age Specific Reproduction

Male insemination was described to behave as a quadratic function with age ($t = -10.86$, $P < 0.001$; Table 1), demonstrating that male insemination success is dependent on age (Fig. 4). Male reproduction exhibited three distinct periods: 1) reproductive onset (ages 0-20), 2) plateau at reproductive maximum (ages 20-35), and 3) apparent reproductive senescence (ages 35-death) (Figure 4). The diet treatments were found to have a significant effect on average insemination success ($F_{2, 40} = 4.07$, $P = 0.024$), with both delayed full diet and sugar diets reducing average insemination success compared with constant full diet ($t = -2.31$, $P = 0.025$; $t = -2.60$, $P = 0.012$ respectively; Table 1). However, there was no difference in the insemination success of males provided the sugar diet and those that experienced a 20 day delay in access to full diet ($t = 0.04$, $P = 0.966$).

Discussion

Female access reduced the life expectancy of male Mexican fruit flies fed a full diet by 30%, a result consistent with the cost of male reproduction reported in other studies on insects (Kotiaho & Simmons, 2003; Martin & Hosken, 2004; Papadopoulos, *et al.*, 2010). Moreover, the effect of female access on was influenced by the age that males first had access to females, so that both truncated and delayed access to females improved the life expectancy of males compared to when females were always available. However, only truncated female access prevented the loss life expectancy due to female access. This result is likely due to males with truncated female access having fewer opportunities to mate, since most Mexican fruit fly males are not expected to be sexually mature until age 10 days (Pereira *et al.* 2013) and reproductive success does not appear to peak until after age 20 days. Delaying female access lessens its cost on male life expectancy because reproduction is postponed until ages when reproductive ability

appears to be declining. Therefore, delaying reproduction limits the males' opportunity to mate until they are older and less likely to copulate.

Sugar diet was detrimental to the lifespan of male Mexican fruit flies, reducing the life expectancy by 40% compared to unpaired males fed the full diet. Surprisingly, the life expectancies of paired and unpaired males fed sugar diet were not different. Initially this result appears to suggest that males are avoiding the cost of female access when only sugar is available, thus supporting the assumptions of optimal resource allocation models (Perrin & Sibly, 1993). However, sugar fed males still mated when provided with females, albeit at reduced success. Such a decrease in insemination success is expected to reduce the cost of reproduction (Kuijper, *et al.*, 2006). Yet, if sugar diet prevented the cost of female access in males, the life expectancy of males with truncated female access should be the same as those of paired and unpaired males. Instead, early life access to females in combination with sugar diet increased male life expectancy. It is unclear from the current study why solitary males fed only sugar have the same life expectancy as the constantly paired males, while the lifespan benefit of remaining unpaired is only manifested for solitary males fed full diet. A possible explanation is that males do not avoid the cost of reproduction when they are fed a poor diet, but the poor quality diet acts as a limiting factor on survival. Therefore, the effect of diet has a stronger influence on male survival than the effect of mating.

Contrary to the effect of dietary timing in female tephritids (Carey *et al.*, 1998), delaying access to full diet does not appear elicit a reproductive waiting period in male Mexican fruit flies when females are not available, so males gain no lifespan advantage when full diet is delayed. Additionally, there was no improvement in male life expectancy from delaying both female access and full diet. Consequently, sugar diet at eclosion nullifies the improvement in life

expectancy associated with delayed female access. The loss of reproductive ability early in the lifespan due to the poor dietary conditions is not recovered through increased age specific reproductive ability following the introduction of full diet, nor is the male lifespan and /or reproductive period extended.

As observed previously, the reproductive ability of male Mexican fruit flies was influenced by diet (Liedo, *et al.*, 2013; Pereira *et al.*, 2013). While the nutrients in the full diet were not required for successful insemination, they were required to maximize copulation success throughout the lifespan. Any period of sugar only diet reduced the overall insemination success of males. This reduction in insemination success may be influenced by the reduced production of certain pheromone compounds, as reported by Liedo *et al.* (2013), which are needed to initiate copulation with the females. Additionally the males may have been weakened or made less attractive to the females by the sugar diet, making them less likely to copulate. The loss of reproductive success due to sugar diet could not be reversed when full diet was provided at age 20 days. Therefore, the lack of appropriate nutrients early in life constrains the male's lifetime mating ability.

The effect of age on reproductive ability was not found to vary among diet types. Regardless of dietary treatment, males demonstrate three distinct age specific reproductive periods corresponding to those observed in female insects (Novoseltsev *et al.*, 2003; Novoseltsev *et al.*, 2004). First was the period of reproductive onset, from age 0-20 days, when the males began to mature. Next, reproductive maturity was reached between ages 25 and 30 days, as demonstrated by the plateau of reproductive success when the highest levels of insemination success were reached. Finally, the period of maturity ended with the onset of reproductive senescence at 30 days, when insemination success began to decline. Physiologically controlled

reproductive senescence is well documented in females from various species (Austad, 2010; Haaga *et al.*, 2010), but sexual selection theory and experimental data suggest that male reproductive performance should increase with age (Williams, 1966; Bonduriansky *et al.*, 2008; Maklakov *et al.*, 2009). Contrary to these predictions, reproductive senescence in the male Mexican fruit fly began when nearly 80% of the males were still alive and the risk of mortality was still relatively low. However, we cannot neglect that the decline in reproduction is a result of behavioural changes instead of a physiologically induced decline in reproductive ability. Male tephritids have been reported to decrease reproductive effort in a response to prior reproductive success and age (Liedo, *et al.*, 2002; Lopez-Guillen *et al.*, 2008), and the willingness of the female tephritid, *Ceratitis capitata*, to mate with males has been observed to be negatively correlated with the male's age (Papanastasiou, *et al.*, 2011). However, these behaviours may be linked to a physiological decline in reproductive ability, and would therefore be mutually affecting, but further investigation is required to isolate the effects of each on age specific reproduction.

Based on these results, male Mexican fruit flies do not gain a reproductive advantage by avoiding reproductive effort when dietary protein and females are unavailable. The males are still able to successfully reproduce in the absence of a protein source, even though the poor dietary conditions shorten their life expectancy. Additionally, any delay in reproductive effort is expected to result in a loss of reproductive opportunity associated with aging, so that the males cannot gain a fitness benefit through delaying reproduction and extending the lifespan. Moreover, this possibility of male reproductive senescence, in which male reproductive ability is affected by both natural lifespan and the onset of senescence, challenges the hypotheses of the

288 male disposable soma theory, in which male reproductive effort should increase with age
289 (Williams, 1966).

290 **Acknowledgements**

291 We thank R.E. Bustamente, S. E. Salgado, E. De Leon, R. Rincon, A. Oropeza
292 S. L. Rodriguez, and G. Rodas for logistical support and assistance in conducting experiments
293 and J.M. Hash, N.T. Papadopoulos, R.I. Vargas, F. G. Zalom, and P. Lower for comments on
294 earlier drafts and editorial assistance. This research was funded through the NIH/NIA PO1
295 programme, the Biodemographic Determinants of Life Span (P01 AG022500-01 and P01
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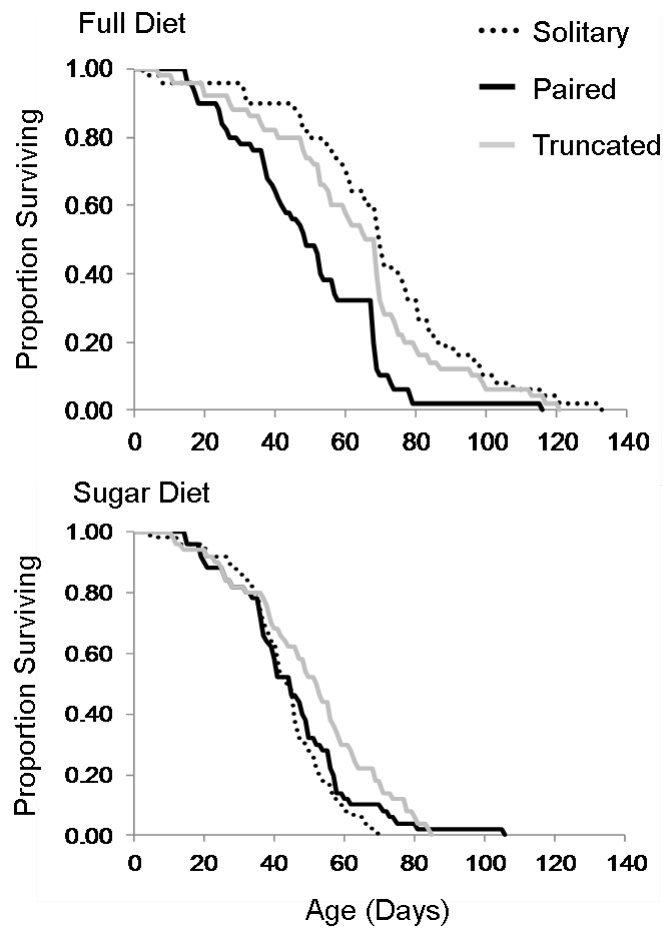


Fig. 1. Age specific survival of adult male Mexican fruit flies when fed either a full diet (3:1 sugar to yeast protein) or sugar only diet, when female access was prevented (solitary), constant from eclosion until death (paired), or only provided until age 20 days (truncated).

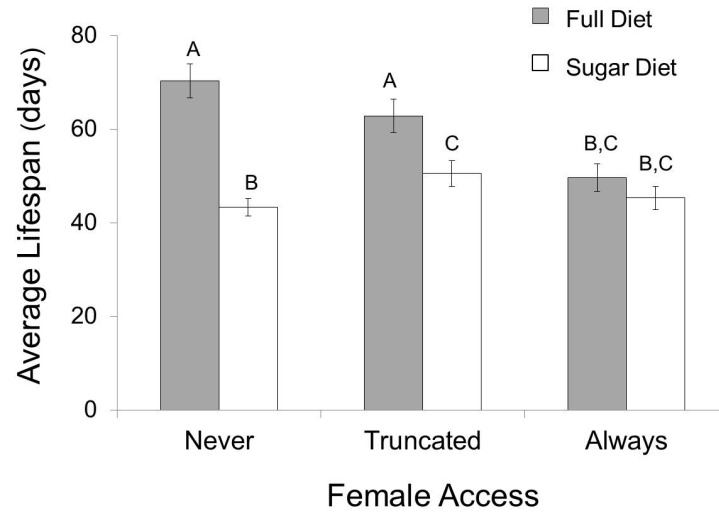


Fig. 2. The average lifespan (days) $\pm$ SE of male Mexican fruit flies provided full diet (3:1 sugar to yeast protein) or a sugar only diet when female access was prevented (never), only provided until age 20 days (truncated), or constant from eclosion until death (always). Averages with different letters represent significant differences ($P < 0.05$) among treatments, as determined through a linear regression and subsequent t-tests.

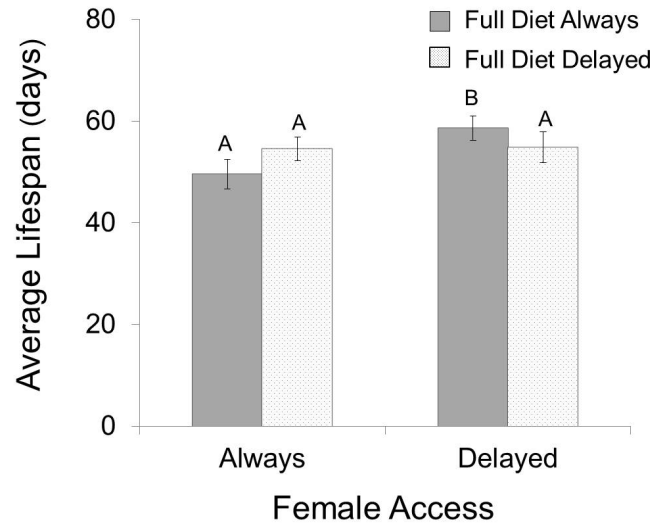
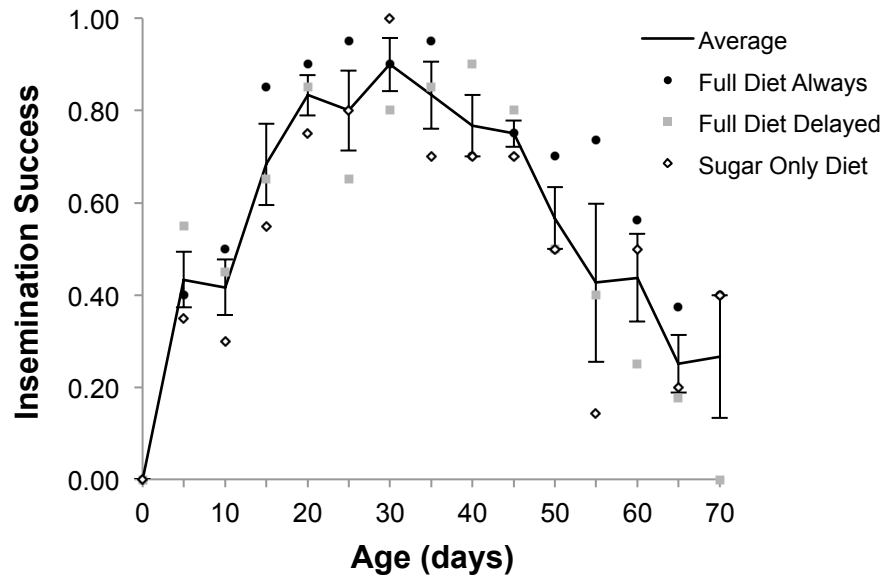


Fig. 3. The average lifespan (days) $\pm$ SE of male Mexican fruit flies provided full diet (3:1 sugar to protein derived yeast) at eclosion (always), or at age 21 days (delayed), prior to which sugar only diet was provided, when females were provided beginning at eclosion (always), or beginning at age 21 days (delayed). Averages with different letters represent significant differences ($P < 0.05$) among treatments, as determined through a linear regression and subsequent t-tests.

425



426

427 **Fig. 4.** Age specific insemination success of male Mexican fruit flies, measured as the proportion
 428 of females to lay eggs that produced larvae, when full diet was provided from eclosion until
 429 death (full diet always), when it was delayed until age 21 days, prior to which sugar diet was
 430 provided (full diet delayed), and when only sugar diet was provided (sugar diet). The line is the
 431 average proportion of males to exhibit successful insemination across all diets at each age with
 432 error bars representing the standard error about the mean at each age.

433

434 **Table 1.** The estimated effect of each diet factor (level) and the covariate Age on the average ($\pm$
 435 SE) proportion of male Mexican fruit flies that successfully inseminated females. The estimate
 436 shows how the diet affects insemination success relative to the intercept (full diet), with the t-
 437 values and p-values calculated from two tail t-tests. Age Squared shows the quadratic effect of
 438 age on insemination success (See Figure 4).

	Estimate	<i>t</i> value	<i>P</i>
Full Diet (Intercept)	0.242 $\pm$ 0.065	3.705	0.001
<u>Diet (level)</u>			
Sugar Only	-0.123 $\pm$ 0.053	-2.313	0.026
Delayed Full Diet	-0.139 $\pm$ 0.053	-2.604	0.013
<u>Covariates</u>			
Age	0.039 $\pm$ 0.004	10.255	>0.001
Age Squared	-0.001 $\pm$ 0.000	-10.863	>0.001

439